METRIC

MIL–PRF–51527B
27 February 2006

SUPERSEDING
MIL–PRF–51527A(EA)
13 March 2000

PERFORMANCE SPECIFICATION

FILTER SET, GAS-PARTICULATE, 200 CFM, M98

Reactivated for New Design after 13 March 2000

This specification is approved for use by the U.S. Army Edgewood Chemical Biological Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of gas-particulate filter set rated at 200 standard cubic feet per minute (scfm) for use in Collective Protection Equipment (CPE) (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to insure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: U.S. Army Edgewood Chemical Biological Center, ATTN: AMSRD–ECB–ENA–S, 5183 Blackhawk Road, Aberdeen Proving Ground, MD 21010–5424 or emailed to SpecsTeam@apgea.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL−DTL−32101	–	Carbon, Activated, Impregnated, Copper-Silver-Zinc-Molybdenum-Triethylenediamine (ASZM-TEDA)
MIL-S-901	–	Shock Test, High-Impact Shipboard Machinery, Equipment, and Systems, Requirements for

DEPARTMENT OF DEFENSE STANDARDS

MIL−STD−130	–	Identification Marking of U.S. Military Property
MIL−STD−167−1	–	Mechanical Vibrations of Shipboard Equipment
MIL−STD−282	–	Military Standard, Filter Units, Protective Clothing, Gas−Mask Components and Related Products: Performance Test Methods
MIL−STD−810	–	Environmental Engineering Considerations and Laboratory Tests
MIL−STD−1168	–	Ammunition Lot Numbering and Ammunition Data Cards

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111−5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

NAVAL SEA COMMAND

Drawing 53711−6263500	–	Filter System, CPS

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SS200−AG−MMM−010 – Navy Shipboard Collective Protection System (CPS) Technical Manual, CBR Filter System Operation and Maintenance

(Copies are available from Naval Surface Warfare Center Dahlgren Division, 17320 Dahlgren Road, Code G52, Dahlgren, VA 22448−5100.)

CODE OF FEDERAL REGULATIONS

40 CFR Part 261 – Identification and Listing of Hazardous Waste

(Copies are available from the U.S. Government Printing Office, Washington, DC 20402 or http://www.access.gpo.gov.)

2.3 Non−Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AIR−CONDITIONING AND REFRIGERATION INSTITUTE (ARI)

ARI 700 – Specifications for Fluorocarbon and Other Refrigerants

(Copies are available from ARI, 4100 N. Fairfax Dr., Suite 200, Arlington, VA 22203 or http://www.ari.org).

AMERICAN CONFERENCE OF GOVERNMENTAL INDUSTRIAL HYGIENISTS

Publication – Threshold Limit Values and Biological Exposure Indices

(Copies are available from ACGIH, 1330 Kemper Meadow Dr., Suite 600, Cincinnati, OH 45240 or http://www.acgih.org.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 209 – Standard Specification for Aluminum and Aluminum Alloy Sheet and Plate

ASTM D 1056 – Standard Specification for Flexible Cellular Materials – Sponge or Expanded Rubber

ASTM D 2867 – Moisture in Activated Carbon

(Copies are available from http://www.astm.org or ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428−2959.)

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SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE AMS 3195 – Silicone Rubber Sponge, Closed Cell, Medium

(Copies are available from SAE, 400 Commonwealth Dr., Warrendale, PA 15096 or http://www.sae.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 12103−1 – Road Vehicles – Test Dust for Filter Evaluation – Part 1

(Copies are available online from http://www.iso.org.)

UNDERWRITERS LABORATORY, INC (UL)

UL 900 – Air Filter Units – Standard for Safety

(Copies are available online from http://www.comm−2000.com.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspections in accordance with 4.2.

3.2 Interface requirements.

3.2.1 Size and shape. The filter set shall be consist of one gas filter and one particulate filter, both rated at 200 scfm. Each gas and particulate filter in the set shall be separable and not permanently attached to each other. The particulate filter shall fit inside the gas filter. Each filter in the set shall be cylindrical in form and configured so that air shall flow through the filter radially outward. Each filter set shall fit easily into its intended housing (see 6.4) and be removed with no damage to itself or the surrounding components. Each shall have strict tolerances for proper housing interface and meet the interface dimensions in Figure 1 and 2, respectively. Each filter end or end cover shall contain one groove for gasket location. The gasket shall be grade number 2C1 of ASTM D1056, or silicone rubber per SAE AMS−3195, be seamless, and have a thickness of 0.787 cm (0.31 in). The gasket shall permanently attached to the filter with an adhesive.

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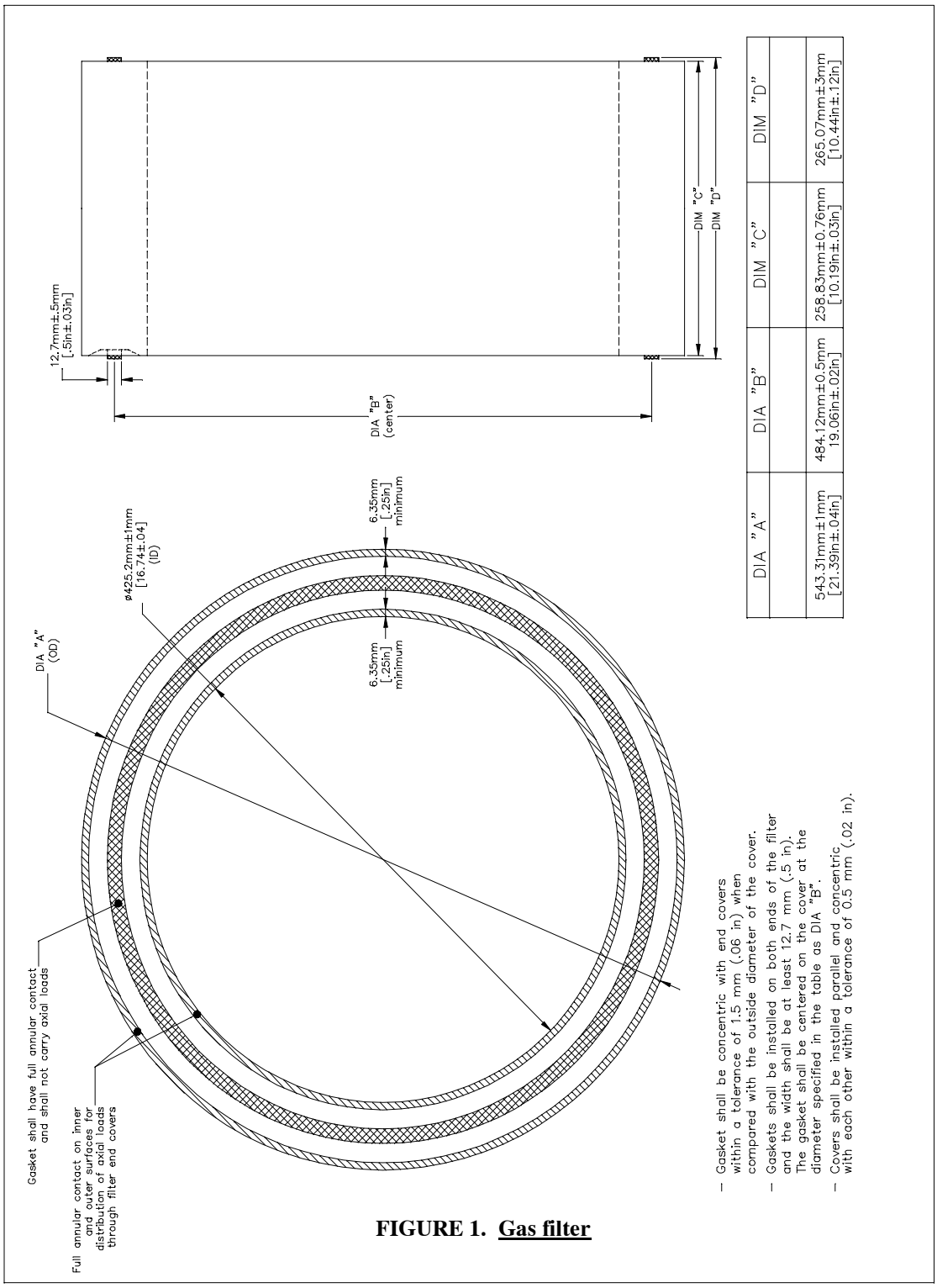


FIGURE 1. Gas filter

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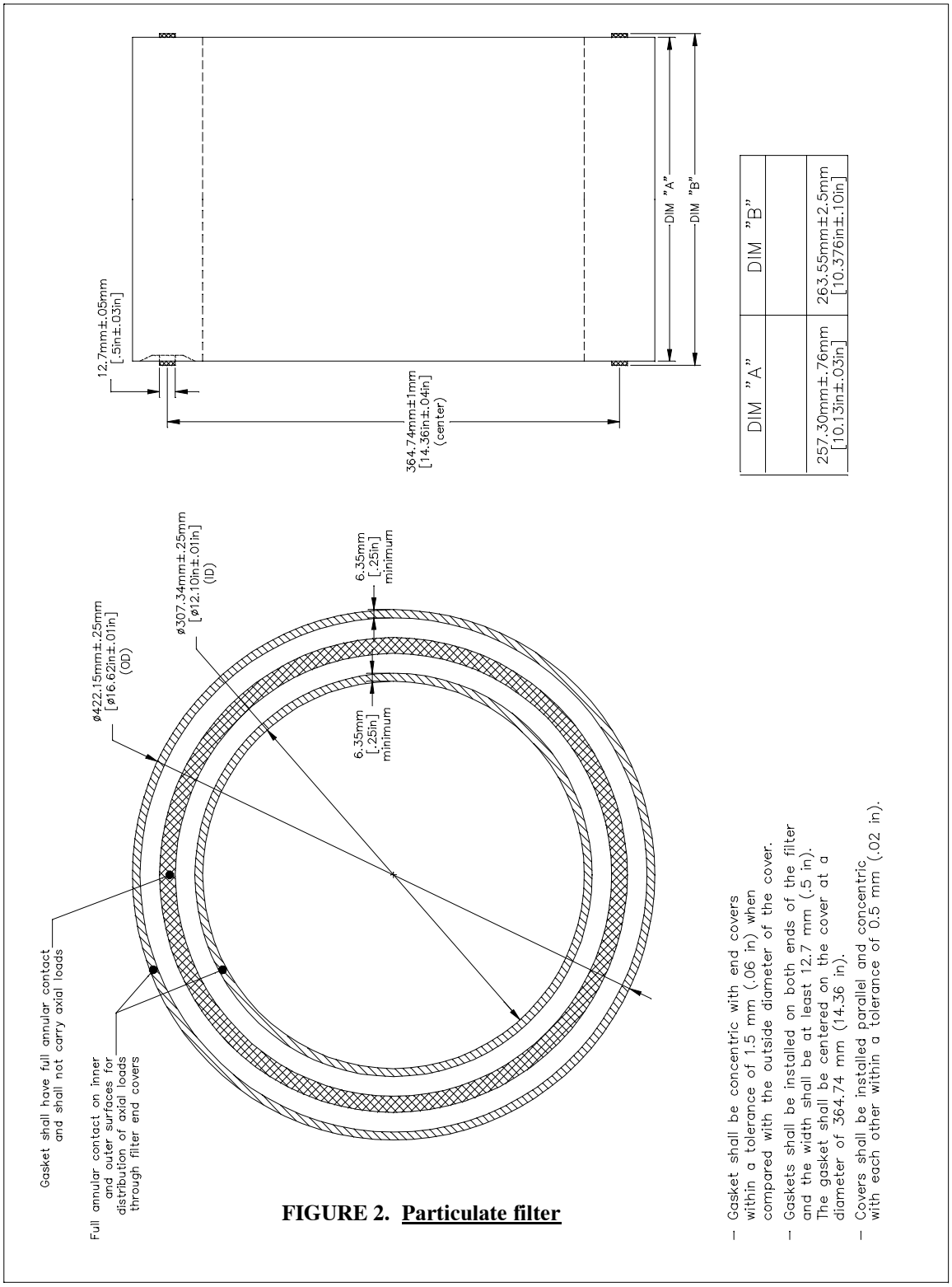


FIGURE 2. Particulate filter

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3.2.2 Weight. The weight of the filter set shall be no greater than 50 pounds while unpackaged.

3.2.3 Interchangeability. Each gas and particulate filter and any associated gasket shall be individually interchangeable (replaceable) by one of similar form, fit and function without modification of the filter housing.

3.3 Operating requirements.

3.3.1 Airflow resistance.

a. The airflow resistance (pressure drop) of the gas filter shall not exceed 4.5 inches of water gage (iwg) at the rated airflow of 200 scfm. Standard conditions is defined as an airstream temperature of 70°F and a barometric pressure of 760 millimeters (mm) Hg.

b. The airflow resistance (pressure drop) of the particulate filter shall not exceed 0.7 inches of water gage (iwg) at the rated airflow of 200 scfm.

3.3.2 Leakage. The gas filter in the filter set shall not leak when a concentration of 1000 parts per million (ppm) of R−134a (1,1,1,2) tetrafluoroethane refrigerant is introduced at the filter inlet.

3.3.3 CK gas life. Each carbon adsorbent sample obtained from the filling process of the filter shall have a cyanogen chloride (CK) life of no less than that specified in MIL−DTL−32101 for unaged carbon.

3.3.4 DMMP gas life. Following filter rough handling, the dimethyl methylphosphonate (DMMP) gas life shall be as follows when the influent concentration is 5000 mg/m^3 at rated flow:

a The mean DMMP gas life minus 1.6 times the standard deviation of the filters shall be no less than 70 minutes.

b. No individual sample filter shall have a DMMP gas life of less than 60 minutes (see 6.9).

3.3.5 Adsorbent media. The adsorbent media in gas filter shall meet all requirements in MIL−DTL−32101 for ASZM−TEDA carbon.

3.3.6 Filtration efficiency. The particulate collection efficiency of 0.3−micron diameter particles shall be at least 99.97 percent when the particulate filter is challenged, at the rated airflow, by an aerosol.

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3.4 Environmental requirements.

3.4.1 Hot/cold temperature.

a. After temperature conditioning at -50 and 160˚F, the gas filter shall show no ignition or structural damage, and shall pass airflow resistance (3.3.1) and DMMP gas life (3.3.4) requirements.

b. After temperature conditioning at −50 and 160˚F, the particulate filter shall show no ignition or structural damage, and shall pass airflow resistance (3.3.1) and filtration efficiency (3.3.6) requirements.

3.4.2 High humidity. After conditioning at 80% relative humidity or above and 113 °F for 7 days, the particulate filter shall pass airflow resistance (3.3.1) and filtration efficiency (3.3.6) requirements.

3.4.3 Structural integrity.

3.4.3.1 Structural deformation.

a. The gas filter shall be capable of withstanding repeated applications of a quasi−static mechanical load of 2,900 pounds in the axial direction. Each gasket shall not become dislodged from the filter by the process of loading a filter into a housing and removing it. After load testing, the filter shall pass airflow resistance (3.3.1) and DMMP gas life (3.3.4) requirements.

b. The particulate filter shall be capable of withstanding repeated applications of a quasi−static mechanical load of 12900 newtons (2900 pounds) in the axial direction. Each gasket shall not become dislodged from the filter by the process of loading a filter into a housing and removing it. After testing, the filter shall meet the filtration efficiency requirement in 3.3.6.

3.4.3.2 Shock.

a. After shock treatment, the gas filter shall pass the DMMP gas life requirement (3.3.4).

b. After shock treatment, the particulate filter shall pass airflow resistance (3.3.1) and the filtration efficiency (3.3.6) requirements.

3.4.3.3 Vibration.

a. After sinusoidal vibration between 4 and 50 Hz, the gas filter shall pass the DMMP gas life requirement (3.3.4).

b. After sinusoidal vibration between 4 and 50 Hz, the particulate filter shall pass airflow resistance (3.3.1) and the filtration efficiency (3.3.6) requirements.

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3.4.3.4 Rough handling.

a. After rough handling, consisting of two hundred 3/4-inch drops per minute for 15 minutes, the gas filter show no evidence of cracks or dents, and shall pass airflow resistance (3.3.1) and DMMP life (3.3.4) requirements.

b. After rough handling, consisting of two hundred 3/4-inch drops per minute for 15 minutes, the particulate filter shall show no evidence of cracks or dents, and shall pass airflow resistance (3.3.1) and filtration efficiency (3.3.6) requirements.

3.4.4 Fungus. The materials of construction, except for ASZM−TEDA carbon, shall not support fungus and mildew growth beyond trace levels.

3.4.5 Corrosion. Any construction material shall resist salt fog corrosion to an extent equal or greater than anodized aluminum when the aluminum alloy number is in the 3000 to 5000 series of ASTM B 209.

3.4.5.1 Coating. {If the material used for the gas filter's housing is metal, its inside surface shall have a coating which prevents galvanic corrosion between the metal housing and the metal impregnants (including any metallic oxides and chlorides) of the carbon sorbent. Coating shall have these characteristics: peel−resistant, electrically insulating, and non−degrading at the temperature and humidity limits in this specification.}

3.4.6 High humidity. After conditioning at 80% relative humidity or above and 113 °F for 7 days, the particulate filter shall pass airflow resistance (3.3.1) and filtration efficiency (3.3.6) requirements.

3.4.7 Dust capacity. The particulate filter shall deliver the rated airflow at a resistance of no greater than 5.7 inch of water when uniformly loaded with 2 kilograms (4.4 pounds) of dust per Class A2 of ISO 12103−1.

3.4.8 Overpressure. The particulate filter shall satisfy the minimum efficiency requirement of 3.3.6 after being subjected to a repeated overpressure that is produced by an airflow of 400 scfm.

3.4.9 Flammability. The particulate filter, when clean, shall not produce flame, sparks or smoke in quantities greater than that allowed for a Class 2 air filter unit per UL 900.

3.5 Ownership and support requirements. These requirements apply to both the gas and particulate filters in the set except for 3.5.2.

3.5.1 Shelf life. Filter set shelf life shall be at least 5 years (extendable) (see 6.8). Shelf life is the maximum time an item may be in sheltered storage within its packaging and still remain fit for use.

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3.5.2 Moisture content. Prior to packaging the gas filter in a sealed container, the moisture content of the carbon shall not exceed 3.0 percent by weight to enhance shelf life.

3.5.3 Safety, health, environment.

3.5.3.1 Materials. The materials of construction shall be non−toxic, non−hazardous, and non−carcinogenic. They shall not be capable of having any adverse effects on human health during normal use. If special use conditions create a health hazard potential, the equipment shall have appropriate hazard warning labels in accordance with 29 CFR 1910.1200.

3.5.3.2 Particulate emissions. The filter set shall pose no potential inhalation hazard to the user, and whose normal use will not exceed the lower of (a) 8−hour time weighted average Threshold Limit Values (TLVs) published by the American Conference of Governmental Industrial Hygienists (ACGIH), or (b) 8−hour time weighted average Permissible Exposure Limits (PELs) regulated by Occupational Safety and Health Administration (OSHA) and found in Material Safety Data Sheets (MSDS) (see 6.7).

3.5.3.3 Odors. When air is forced through the filter at 200 scfm, filter set shall not emit an irritating level of particulate emissions or obnoxious odors.

3.5.3.4 Hazardous waste. At the time of disposal, uncontaminated filter sets (including any coatings) shall not be a Resource Conservation Recovery Act hazardous waste according to characteristics cited in 40 CFR 261.21−261.24 and shall not be listed as specific hazardous waste chemicals in 40 CFR 261.33(e) and (f).

3.5.3.5 Sharp edges. Filter sets shall be free from burrs and sharp edges that might injure personnel, damage filter media, or damage Mission Oriented Protective Posture (MOPP) IV gloves during handling.

3.5.4 Identification markings. All markings shall be accessible, legible and permanent.

a. The gas filter shall contain markings on the outside surface which include the name “FILTER, GAS, 200 CFM”, contract number, lot number, serial number, national stock number, cage code, manufacturer’s name, and manufacture date. Letters shall be at least 0.2 inch high.

b. The particulate filter shall contain markings on the outside surface which include the name “FILTER, PARTICULATE, 200 CFM”, contract number, lot number, serial number, national stock number, cage code, manufacturer’s name, and manufacture date. Letters shall be at least 0.2 inch high.

c. Each filter shall also contain a barcode from which the following may be read: name, contract number, lot number, serial number, national stock number, cage code, manufacturer’s part number, manufacture date, expiration date, and measured airflow resistance and penetration percent. Barcode dimensions and characteristics shall comply with MIL-STD-130.

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3.5.5 Workmanship.

a. The filter sets shall be free from foreign matter (dirt, oil, or viscous materials).

b. The filter sets shall be free from cracked, bent, or dented metal sections, and abraded gaskets.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)
b. Conformance inspection (see 4.3)

4.2 First article inspection.

4.2.1 Sample. The first article sample shall consist of 20 gas−particulate filter sets manufactured using the same methods, materials, equipment, and processes as will be used during regular production. The first article sample shall be submitted for inspection and approval in accordance with the terms of the contract.

4.2.2 Inspections to be performed. Unless otherwise specified by the terms of the contract, the sample first article items shall be subjected to all of the verifications specified in this specification. Verification of requirements shall be by examination, (examination includes visual inspection with or without simple commercial measuring devices), demonstration, test data, or by Certificate of Compliance (COC) with analysis or supporting evidence. The Government reserves the right to accept or reject each COC. First article filters shall be inspected according to Figures 3 and 4, or as determined by the Government (see 6.2). Each filter shall be examined for interface characteristics, identification markings, workmanship, and safety hazards. Each filter shall then be tested for airflow resistance. Each gas filter shall be tested for leakage and each particulate filter shall be tested for filtration efficiency. Following moisture content testing of adsorbent material, sample gas filters shall be packaged using the same methods, materials, equipment, and processes as will be used during regular production. Packaged filters shall then be forwarded to the Government for rough handling and additional tests. Any packaged sample filters forwarded to the Government shall be protected from moisture in the air. Requirements and verification methods are cross referenced in Table I.

4.2.3 Acceptance criteria. If any first article sample item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

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TABLE I. First article inspection

Description	Requirement Paragraph	Verification Paragraph
Interface requirements	3.2	4.4.1
Size and shape	3.2.1	4.4.1.1
Weight	3.2.2	4.4.1.2
Interchangeability	3.2.3	4.4.1.3
Operating requirements	3.3	4.4.2
Airflow resistance	3.3.1	4.4.2.1
Leakage	3.3.2	4.4.2.2
CK gas life	3.3.3	4.4.2.3
DMMP gas life	3.3.4	4.4.2.4
Adsorbent media	3.3.5	4.4.2.5
Filtration efficiency	3.3.6	4.4.2.6
Environmental requirements	3.4	4.4.3
Hot/cold temperature	3.4.1	4.4.3.1
High humidity	3.4.2	4.4.3.2
Structural integrity	3.4.3	4.4.3.3
Structural deformation	3.4.3.1	4.4.3.3.1
Shock	3.4.3.2	4.4.3.3.2
Vibration	3.4.3.3	4.4.3.3.3
Rough handling	3.4.3.4	4.4.3.3.4
Fungus	3.4.4	4.4.3.4
Corrosion	3.4.5	4.4.3.5
High humidity	3.4.6	4.4.3.6
Dust capacity	3.4.7	4.4.3.7
Overpressure	3.4.8	4.4.3.8
Flammability	3.4.9	4.4.3.9
Ownership and support	3.5	4.4.4
Shelf life	3.5.1	4.4.4.1
Moisture content	3.5.2	4.4.4.2
Safety, health, environment	3.5.3	4.4.4.3
Identification markings	3.5.4	4.4.4.4
Workmanship	3.5.5	4.4.4.5

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<table>
<tr><td colspan="7">First article sample (19 gas filters plus one spare)</td></tr>
<tr><td colspan="7">Contractor's quality inspections including leakage tesing</td></tr>
<tr><td colspan="7">Filters shipped to Government</td></tr>
<tr><td colspan="7">Interface, identification markings, workmanship
Airflow resistance</td></tr>
<tr><td>Shock

(Qty 3)</td><td>Vibration

(Qty 3)</td><td>Rough handling

(Qty 6)</td><td>Hot/cold temperature

(Qty 2)</td><td>Structural deforma-tion
(Qty 2)</td><td>Corrosion (salt fog)

(Qty 1)</td><td>Emission test & analysis
(Qty 2)</td></tr>
<tr><td colspan="7">Airflow resistance (Qty 19)
DMMP gas life (Qty 19)</td></tr>
</table>

FIGURE 3. First article inspection sequence, gas filters

<table>
<tr><td colspan="7">First article sample (20 particulate filters)</td></tr>
<tr><td colspan="7">Contractor's quality inspections</td></tr>
<tr><td colspan="7">Filters shipped to Government</td></tr>
<tr><td colspan="7">Interface, identification markings, workmanship, safety
Airflow resistance
Filtration efficiency</td></tr>
<tr><td>Shock

(Qty 3)</td><td>Vibration

(Qty 3)</td><td>Rough handling

(Qty 6)</td><td>Hot/cold temperature,

(Qty 2)</td><td>Structural deforma-tion
(Qty 2)</td><td>Over−pressure

(Qty 2)</td><td>Dust capacity

(Qty 2)</td></tr>
<tr><td></td><td></td><td></td><td>High humidity
(Qty 2)</td><td></td><td></td><td></td></tr>
<tr><td colspan="4">Airflow resistance (Qty 14)
Filtration efficiency (Qty 14)</td><td colspan="3">Filtration efficiency (Qty 6)</td></tr>
<tr><td colspan="7">Corrosion (salt fog) (Qty 2)
Flammability (Qty 2)</td></tr>
</table>

FIGURE 4. First article inspection sequence, particulate filters

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4.3 Conformance inspection.

4.3.1 Lotting. A lot shall consist of a series of filter sets whose separate gas and particulate filters are each produced by one manufacturer, at one plant, from the same materials, under the same manufacturing and process conditions and without a break in production of more than 10 calendar days. The gas and particulate filters in a given lot of filter sets need not be produced from the same manufacturer. Any one lot of gas filters shall contain no more than one lot of adsorbent media. Each lot shall be identified and controlled in accordance with MIL−STD−1168.

4.3.2 Sampling. Inspection and test of random samples shall be conducted in accordance with the classification of characteristics in 4.3.5, and when specified in Table II.

4.3.3 Inspection procedure. Every item in the lot shall be inspected for critical characteristics. Sample filters shall be examined and tested in accordance with the classification of characteristics in 4.3.5. If a filter is found that does not conform to any characteristic inspected 100%, such as pressure drop requirements, the non−conforming filter shall be rejected and removed from the lot. For a characteristic not inspected 100% in the classification of characteristics table, failure to conform to the sampling and acceptance criteria specified therein shall be cause for rejection of the lot represented (see footnote, Table II).

4.3.4 Inspection characteristics. Critical characteristics are characteristics whose nonconformance to specified requirements is likely to result in hazardous or unsafe conditions for individuals who use or maintain the product. Characteristics whose nonconformance to specified requirements is likely to prevent performance of the tactical function of a major end item are also critical characteristics. Major characteristics are characteristics whose nonconformance to specified requirements is likely to result in failure or to reduce materially the usability of the item for its intended purpose. Minor characteristics are characteristics whose nonconformance to specified requirements is not likely to reduce materially the operation or usability of the item for its intended purpose.

4.3.5 Classification of characteristics. Conformance inspections shall be as specified in Table III classification of characteristics. When specified herein, accept on “0” and reject on “1” attributes sampling inspection shall be performed on the designated characteristics using the stated levels in Table II for selection of sample sizes.

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TABLE II. Sampling

Lot size	Inspection levels and sample sizes										
	I	II	III	IV	V	VI	VII	VIII	IX	X	XI
2 to 8	*	*	*	*	*	*	*	*	5	3	2
9 to 15	*	*	*	*	*	*	13	8	5	3	2
16 to 25	*	*	*	*	*	20	13	8	5	3	3
26 to 50	*	*	*	*	32	20	13	8	5	5	5
51 to 90	*	*	*	50	32	20	13	8	7	6	5
91 to 150	*	*	125	50	32	20	13	12	11	7	6
151 to 280	*	*	125	50	32	20	20	19	13	10	7
281 to 500	*	315	125	50	48	47	29	21	16	11	9
501 to 1200	*	315	125	75	73	47	34	27	19	15	11
1201 to 3200	1250	315	125	116	73	53	42	35	23	18	13
3201 to 10000	1250	315	192	116	86	68	50	38	29	22	15
10001 to 35000	1250	315	294	135	108	77	60	46	35	29	15
35001 to 150000	1250	490	294	170	123	96	74	56	40	29	15
150001 to 500000	1250	715	345	200	156	119	90	64	40	29	15
500001 and over	1250	715	435	244	189	143	102	64	40	29	15

*Indicates one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection.
Accept the lot represented on zero nonconforming characteristics and reject the lot represented on one or more nonconforming characteristics for all inspection levels.

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TABLE III. Classification of characteristics

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.3.5(a)	Filter, Gas, 200 CFM		NEXT HIGHER ASSY

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical				
1	Leakage	100 percent	3.3.2	4.4.2.2
2	CK gas life	See 4.4.2.3.2	3.3.3	4.4.2.3
3	DMMP gas life	Table II, level XI*	3.3.4	4.4.2.4
Major				
101	Airflow resistance	100 percent	3.3.1	4.4.2.1
102	Moisture content	See 4.4.4.2.2	3.5.2	4.4.4.2
Minor				
201	Identification markings	Table II, level VII	3.5.4	4.4.4.4
202	Workmanship	Table II, level VII	3.5.5	4.4.4.5

NOTES:

* Except no lower than five samples.

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TABLE III. Classification of characteristics (continued)

PARAGRAPH	TITLE		DRAWING NUMBER
4.3.5(b)	Filter, Particulate, 200 CFM	SHEET 1 OF 1	

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	Filtration efficiency	100%	3.3.6	4.4.2.6
Major				
101	Airflow resistance	100%	3.3.1	4.4.2.1
102	Over−pressure and filtration efficiency	Table II, level IX	3.4.7 3.3.6	4.4.3.7 4.4.2.6
103	Safety, environment and health	Table II, level IX	3.5.3	4.4.4.3
Minor				
201	Identification markings	Table II, level IX	3.5.4	4.4.4.4
202	Workmanship	Table II, level IX	3.5.5	4.4.4.5

NOTES:

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4.4 Verification methods and procedures.

4.4.1 Interface verification.

4.4.1.1 Size and shape. Verify by examination and measurement using commercial inspection equipment.

4.4.1.2 Weight. Verify by examination and measurement using commercial inspection equipment.

4.4.1.3 Interchangeability. Verify by certification and analysis of filter design and surrounding interfacing hardware.

4.4.2 Operating verification.

4.4.2.1 Airflow resistance. The airflow resistance shall be determined at the rated airflow of 200 scfm. (Air shall flow through the filter radially outward). The measured pressure drop across the filter, when corrected to standard conditions of 70˚F and 1 atm (760 mm Hg), shall be no greater than that specified in 3.3.1. The up−stream and down−stream static pressure measuring tubes shall be as close as possible to the filter and shall not be on a section of duct that has a changing cross sectional area. Test for airflow resistance as follows:

a. Connect the filter to a source of forced air.
b. Set the flow of air through the filter to 200±10 cfm.
c. Record the barometric pressure.
d. Measure and record the air stream temperature. This temperature should be no higher than 125˚F.
e. Determine and record the difference between up-stream and downstream static pressure.

If the recorded air stream temperature and barometric pressure is different than 70˚F and 1 atm (760 mm Hg), calculate and record the air flow resistance (ΔP (cal)) using the equation below to correct the measured airflow resistance to airflow resistance at standard conditions:

$$\Delta P\ (cal) = \Delta P\ (measured) \times P\ (test) \times 86.21 / [(460 + T)^{1.768}]$$

where:

ΔP (cal) – air flow resistance corrected to standard conditions in inches of water gage (iwg)
P (test) – barometric pressure at time of test in millimeters of mercury (mm Hg)
ΔP (measured) – air flow resistance from test measurement in iwg
T – temperature of air stream flowing through the filter in degrees F

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Note: Correction for nonstandard conditions is not required when it is clear that the passage or failure of the filter is not in doubt.

4.4.2.2 Leakage. Rigidly position the sample gas filter in the test fixture. Orient the filter in the fixture with the cylindrical axis in the horizontal direction and end faces in the vertical direction. Use R−134a refrigerant that conforms to the characteristics of ARI STANDARD 700. Connect the filter to a source of forced air and place a mixing chamber at the influent side. Regulate the flow of air from the blower to 1/5th the rated airflow of the filter (40 cfm). Introduce R−134a into the intake of the blower; monitor and maintain a concentration of 1000 ppm by volume of R−134a in airflow of 40 cfm on the influent side of the filter at 75 ±9°F and relative humidity of no higher than 50 percent. Leakage shall be determined using a suitable leak detector (see 6.5) for sampling and detecting R−134a at the effluent side of the filter. Monitor the effluent airstream for at least 2 minutes and record the highest effluent penetration percent during that period. Determine whether the effluent concentration exceeds 1 ppm {.5 ppm} within 2 minutes after the introduction of the refrigerant at the inlet of the filter. Purge the filter by passing clean air in the reverse direction through the filter at rated flow for 3 minutes. Keep the exposure of the filter to air to a minimum.

4.4.2.3 CK gas life. Verify by sampling and testing as described below.

4.4.2.3.1 First article. A sample of adsorbent shall be taken during the filling of each of 8 first article filters, placed in an open container, and exposed to the same conditions of temperature and humidity as its associated filter. Each adsorbent sample taken during the filling of those filters shall weigh at least 100 grams (0.22 pounds). The 8 samples shall then be sealed in their containers and sent to the Government. At least 4 of the samples shall be tested for CK life. Perform CK life testing on each unaged carbon sample placed in tubes in accordance with procedures, conditions and parameters in MIL−DTL−32101. Compare CK gas life of each individual specimen and their average value against the required values specified in MIL−DTL−32101.

4.4.2.3.2 Conformance. For each lot, 8 samples of adsorbent, each sample weighing at least 100 grams, shall be taken at random times during the filling of 8 respective filters and placed in an open container. Those samples shall be exposed to the same conditions of temperature and humidity as their associated filters. The 8 samples shall then be sealed in their containers and sent to the Government. At least 4 of the samples shall be tested for CK life. Compare CK gas life of each individual specimen and their average value against the required values specified in MIL−DTL−32101.

4.4.2.4 DMMP gas life. After gas filters have been rough handled or exposed to shock and vibration (see figure 2), they shall be tested for airflow resistance and DMMP gas life. Test for DMMP gas life as follows:

a. Place the test filter in the test apparatus with the filter in its normal service orientation (cylindrical axis in the horizontal position). Switch airflow control to the “purge” mode and pass

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200 scfm of clean, 125°F air through the test filter for approximately 30 minutes or until effluent conditions are stable.

b. Start DMMP test. Switch airflow control to "test" mode, turn DMMP pump on (pump control pre−set to 5 mg/l) and start timer. Maintain the tester at the proper air flow and temperature.

c. Determine the DMMP gas life (the total time from the start of the DMMP challenge until the breakthrough concentration is reached) of the filter at the following test conditions:

Airflow rate: 200±10 scfm
Challenge concentration* of DMMP: 5000±400 mg/m^3
Breakthrough concentration of DMMP: 0.04 mg/m^3
Relative humidity: less than or equal to 50%
Airstream and filter temperature: 120±5°F

*The DMMP challenge concentration in the air shall be monitored to ensure uniform mixing of the DMMP vapor with the air once it reaches the test item.

d. While the challenge concentration is permitted to vary within the stated range during the course of the test, the breakthrough time measured must be normalized to the 5000 mg/m^3 concentration by the following calculation:

$$\text{Corrected breakthrough time} = \text{Measured breakthrough time} * \frac{\text{Average concentration during test}}{5000\ \text{mg/m}^3}$$

4.4.2.5 Adsorbent media. Verify adsorbent media complies with the physical and chemical characteristics in MIL−DTL−32101 by testing according to that specification or provide certificate of compliance with supporting evidence.

4.4.2.6 Filtration efficiency.

4.4.2.6.1 Methodology. Filtration efficiency shall be measured for each sample particulate filter by using Method A or Method B using aerosolized polyalphaolefin as the simulant challenge aerosol (see 6.10.4). In Method A, the filter shall be challenged with a polydispersed aerosol whose count mean diameter is 0.3 microns and whose particle size standard deviation is no greater than 2.0. Method A shall use a particle analyzer for size and count determinations. Mass mean particle size should be measured using one first article filter for comparison with the count mean particle size. In Method B, the filter shall be challenged with a monodispersed or near monodispersed aerosol whose mass mean diameter is 0.3 microns and whose particle size standard deviation is no greater than 1.3. Method B shall use a photometer to measure aerosol concentrations. If particle size test data are log normally distributed, than geometric mean and standard deviation should be computed instead of arithmetic mean and standard deviation.

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4.4.2.6.2 Apparatus. For either method, the test apparatus shall be capable of maintaining a stable concentration of aerosol with a steady flow through the filter of 200 scfm, accurately determining the penetration percent within ± 0.005%, and producing no adverse effects on human health. The aerosol selected should not react with any of the construction materials. The measurement equipment must have an indicator that alerts the user when the equipment is in need of calibration or maintenance. The downstream sample point shall be located such that changing of its location across the ductwork does not cause a significant change in the measured concentrations. This shall be verified using a filter with a known measurable leak.

4.4.2.6.3 Procedure. The particulate filter shall be placed in the test apparatus conditioned at room temperature. Maintain the airflow within 10% of 200 scfm with air flowing radially outward through filter. That airflow shall be maintained while filtration efficiency is recorded. Challenge the filter at a particle count concentration (Method A) or aerosol concentration (Method B) that is within the acceptable range of the equipment manufacturer's specification. If using Method B, challenge the filter at an aerosol concentration of $100 \pm 20\ \mu g/L$. An inlet concentration lower than $100 \pm 20\ \mu g/L$ may be used during conformance testing if the range is within the manufacturer's specification. Filtration efficiency shall be calculated from the particle counts or concentrations as follows:

Penetration % = N_d/N_u * 100 for Method A
Penetration % = C_d/C_u * 100 for Method B

Filtration efficiency % = 100 – penetration %, where

C_d = Concentration downstream of the filter
C_u = Concentration upstream of the filter
N_d = Number of particles downstream of the filter
N_u = Number of particles upstream of the filter

4.4.3 Environmental verification.

4.4.3.1 Hot/cold temperature. Condition each gas and particulate filter for 8 hours at −50˚F in an environmental chamber. Remove filters from chamber and examine them for damage at room temperature. When temperature of filter sets returns to 70 ±10 ˚F, check airflow resistance of each filter. Then, condition each gas and particulate filter for one week at 160˚F in an environmental chamber. Remove filters from chamber and examine them for damage at room temperature. When temperature of filters returns to 70 ±10 ˚F, test each gas filter for airflow resistance and DMMP gas life; test each particulate filter for airflow resistance and filtration efficiency.

4.4.3.2 High humidity. Expose the particulate filter to 80% relative humidity at 84˚F for 7 days. After removing filter from environmental chamber, perform airflow resistance and efficiency testing.

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4.4.3.3 Structural integrity.

4.4.3.3.1 Structural deformation. Each gas and particulate filter shall be subjected to a quasi−static mechanical load of 2900 pounds in the axial direction for 3 cycles of one minute per cycle. Then test filters for airflow resistance and DMMP gas life.

4.4.3.3.2 Shock. Shock testing shall be performed on filter sets that are fully loaded in the U.S. Navy CPS Filter System (53711−6263500). The CPS system has a flow capacity of 600 cfm. Filter sets shall be installed in accordance with the operation and maintenance manual, SS200−AG−MMM−010. The filter housing shall be filled to capacity. The filter system (housing with filter sets) shall be shock tested per MIL−S−901 using an approved mounting fixture. Shock test procedures shall be based on Class I (no resilient mountings), Type B (subsidiary component), and medium weight category defined in MIL−S−901.

4.4.3.3.3 Vibration. Vibration testing shall be performed on filter sets that are loaded in the U.S. Navy CPS Filter System (53711−6263500). The CPS system has a flow capacity of 600 cfm. Filter sets shall be installed in accordance with the operation and maintenance manual, SS200−AG−MMM−010. The filter housing shall be filled to capacity. The filter system (housing with filter sets) shall be vibration tested per MIL−STD−167−1 using an approved mounting fixture. Test procedure shall be based on MIL−STD−167−1, Type I− Environmental Vibration, 4 Hz to 50 Hz, and consist of 3 parts : (1) Exploratory test to determine resonances (2) Variable frequency test (4 to 50 Hz with 5 minute dwell time per hertz), and (3) Two−hour endurance test for each of the 3 orthogonal axes.

4.4.3.3.4 Rough handling. Perform rough handling testing on separate gas and particulate filters that have not previously undergone shock and vibration testing. All sample filters shall be mounted with the cylindrical axis in a vertical orientation and parallel to the direction of movement. No filter shall be allowed to touch any other filter. Filters shall be rough handled in accordance with Method 105.11 of MIL−STD−282. The rough handling shall consist of 200 3/4−inch drops per minute for 15 minutes. Following rough handling, examine bottom plate and surrounding area for evidence of loose carbon dust. Test each gas filter for airflow resistance and DMMP gas life; test each particulate filter for airflow resistance and filtration efficiency.

4.4.3.4 Fungus. Verify, if possible, by analysis of technical documentation such as specifications, design data, existing test reports, and operating experience. Otherwise, verify requirement by performing a fungus test in accordance with Method 508.5 of MIL−STD−810.

4.4.3.5 Corrosion. The manufacturer shall provide objective evidence and certify that any metallic construction materials will resist salt fog corrosion. Objective evidence includes test data or past service life data. If objective evidence does not exist or is disapproved by the Government, perform the following tests on separate gas and particulate filters: conduct salt fog testing in accordance with Method 509.3 of MIL−STD−810 except the total duration shall be 20 days corresponding to five 96−hour exposure cycles. Each 96−hour exposure cycle is defined as 48 hours of salt fog challenge followed by 48 hours of storage at ambient laboratory

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conditions with no salt fog challenge. After the five 96−hour exposure cycles, rinse the external surfaces with clear water if desired, dry, and inspect filter (inside and outside) for evidence of damage and corrosion. If there is evidence of corrosion, test each gas filter for airflow resistance and DMMP gas life; test each particulate filter for airflow resistance and filtration efficiency.

4.4.3.5.1 Coating. {Verify coating meets all characteristics by engineering analysis and certification.}

4.4.3.6 Dust capacity. Prior to conducting the dust capacity test, the particulate filter shall be conditioned for a period of at least 24 hr at ambient laboratory temperature and not more than 50% relative humidity. The dust capacity test apparatus may conform to SAE J726. The apparatus shall be capable of dispersing a dust per grade A2 of ISO 12103−1. Provisions shall be made to determine the total weight of dust injected and for measuring the filter inlet pressure head. The preconditioned filter shall be placed in the test apparatus so that air flows radially outward through the filter during testing. The test shall consist of an initialization phase and a dust injection phase. During the initialization phase, flow through the filter will be adjusted to the rated flow without dust injection. During the dust injection phase, dust will be injected into the inlet air stream at the rate required to establish a dust concentration 1.0 ±0.1 gm/m^3 at the rated airflow under ambient laboratory conditions. Record the inlet head values every 15 minutes after the initiation of the dust injection. The test shall be run until the inlet head reaches or exceeds the resistance value stated in 5.7 iwg, or for 6 hours, or until a mechanical failure is indicated, whichever occurs first. Upon completion of the required dust injection phase, record the total weight of dust injected, and test the filter for filtration efficiency. Any mechanical failure of the filter, such as a sudden surge of flow and/or drop in the inlet head pressure during the dust capacity test phase, or failure of the subsequent efficiency test shall be cause for rejection.

4.4.3.7 Overpressure. Subject the particulate filter to a differential pressure produced by the airflow rate of 400± 20 scfm. After the filter inlet is subjected to this overpressure for 5 minutes, the flow shall be stopped for one minute. This test cycle shall be repeated 3 times. After completion of the 4 overpressure cycles, perform efficiency testing at the rated 200 scfm airflow.

4.4.3.8 Flammability. The combustibility and smoke generating potential of the particulate filter shall be determined by using the apparatus and procedures specified in section 5 and 6 of UL 900. The quantity of flame, sparks and smoke should be evaluated in accordance with procedures in UL 900.

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4.4.4 Ownership and support verification.

4.4.4.1 Shelf life. Verify shelf life by analysis (engineering evaluation) of materials of construction, packaging, surveillance data if available, and the quantity of moisture in the carbon sorbent. Assume storage temperatures up to 160˚F and relative humidities up to 100%. The shelf life estimate should also consider the likelihood and extent of carbon settling with time for both the vertical and horizontal filter orientation.

4.4.4.2 Moisture content. Verify by sampling and testing as described below.

4.4.4.2.1 First article. A sample of carbon adsorbent shall be taken during the filling of the first, middle, and last filter in the first article sample lot. Each carbon sample taken during the filling of those 3 filters shall weigh at least 100 grams (0.22 pounds). Each sample shall be placed in an open container whose diameter is such that the depth of the carbon will be 1.5 ± 0.5 inches deep. Those samples of carbon shall be exposed to the same conditions of temperature and humidity as their associated filters and be used for subsequent moisture testing. Concurrent with or immediately following the packaging of the filters corresponding to each carbon sample, each of 3 carbon samples shall be tested for moisture content in accordance with ASTM D 2867, Oven−Drying Test Method, except that the oven temperature shall be 217 to 224˚F (103 to 107˚C) and the drying time shall be 3 hours. If the moisture content of the adsorbent exceeds the requirement, the first article lot of filters represented by the samples shall be rejected.

4.4.4.2.2 Conformance. A sample of carbon adsorbent shall be taken during the filling of the first and last filter manufactured each day. Each carbon sample taken during the filling of those 2 filters shall weigh at least 100 grams. Each sample shall be placed in an open container whose diameter is such that the depth of the carbon will be 1.5 ± 0.5 inches deep. Those samples of carbon shall be exposed to the same conditions of temperature and humidity as their associated filters and be used for subsequent moisture testing. Concurrent with or immediately following the packaging of the filters corresponding to each carbon sample, each carbon sample shall be tested for moisture content in accordance with ASTM D 2867, Oven−Drying Test Method, except that the oven temperature shall be 217 to 224˚F (103 to 107˚C) and the drying time shall be 3 hours. If the moisture content of the adsorbent exceeds the requirement, the production run of filters fabricated that day shall be rejected.

4.4.4.3 Safety, health, and environment.

4.4.4.3.1 Materials. Verify no adverse effects on health by health hazard assessment or analysis. In the assessment, consider the probability and consequences of failing to meet each requirement. If warning labels are justified, examine the labels for legibility and permanence.

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4.4.4.3.2 Particulate emissions. Verify no inhalation hazard by measuring particulate emission concentration downstream of the gas filter while air passes through a vibrating filter. This test simulates use in a mobile shelter or vehicle. Pass 200 scfm of forced air through the gas filter. Concurrently, during 6 of the 8 operating hours, vibrate filter for 2 hours in each of 3 orthogonal axes according to MIL−STD−810, method 514.5, procedure I−general vibration, figure 514.5C−3 to simulate vehicle vibration. During the 8−hour operating period, pass the entire 200 scfm of air through the gas filter and through a High Efficiency Particulate Filter (HEPA) (not the same HEPA filter that is part of the M98 filter set). Monitor the airflow at least once every hour. Collect any particulate matter collected on the HEPA filter, weigh the amount collected immediately after the first 10 minutes of 200 scfm operation and immediately after 8 hours of operation. Analyze the collected material for the weight percentage of carbon. For operating times of 10 minutes and 8 hours, compare the measured time−averaged carbon concentration against the carbon inhalation threshold of 5 mg/m^3 concentration limit (see 6.7). If the 8−hour time averaged carbon concentration exceeds 5 mg/m^3 , the sample filters shall be rejected. The following alternate method may be used without using a HEPA filter: monitor the total dust concentration at 10 minutes and at 8 hours of operation using a commercial inhalable dust sampler which meets ACGIH sampling criteria.

4.4.4.3.3 Odors. Verify by using human subjects who inhale chamber air for the first hour during particulate emission test. Alternatively, verify by COC based on existing human factors data collected on filters of the same design that are used in shelters.

4.4.4.3.4 Hazardous waste. Verify by certification based on analysis of construction materials and coatings. Certify that all construction materials including any CARC primer coatings are free of hexavalent chromium and do not qualify as hazardous waste as defined by the the Code of Federal Register.

4.4.4.3.5 Sharp edges. Verify by examination of edges on external surfaces.

4.4.4.4 Identification markings. Verify by examination.

4.4.4.5 Workmanship. Verify by examination.

5. PACKAGING

5.1 Packaging. For acquisition purposes, packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DOD personnel or in−house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department or Defense Agency's automated packaging files, CD−ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The gas−particulate filter set is intended for use as part of a Nuclear Biological Chemical (NBC) air handling system of a collective protection system. The filter set(s) may be used in shipboard Collective Protection Systems (CPS), in land−based CPE such as the Modular CPE and Simplified CPE, and in the fan & filter housing of the transportable CPS. Filters are normally mounted with the axis in the horizontal position.

6.2 Acquisition requirements. Acquistion documents should specify the following:

a. Title, number, and date of this specification.
b. First article:
    1. Time allowed for contractor submission of samples for Government test and evaluation after award of contract when testing is performed by the Govern ment.
    2. Name and address of test facilities and shipping instructions when testing is performed by the Government (see 6.3).
    3. Time required for the Government to notify the contractor whether or not to proceed with production.
c. Packaging requirements.
d. Data requirements (e.g. health hazard or risk assessment reports)
e. Marking requirements.
f. Whether any Government Furnished Equipment is supplied.
g. Label requirements. Adhesive−backed labels may need to be purchased so that user may record filter installation date.
h. Whether objectives (desired characteristics) enclosed in braces in section 3 and 4 are mandatory.

6.3 Test facilities. Suggested shipping addresses for Government portion of First Article test are as follows:

a. For navy−type shock and vibration testing:
    NU Laboratories, Inc.
    312 Old Allerton Road
    Annandale, NJ 08801
    (908) 713−9300

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b. For receipt, examination, and distribution to test activities at ECBC:
U.S. Army ECBC
AMSRD−ECB−ENE−M
Building E5165
APG, MD 21010−5424

6.4 Technical data. These specifications and drawings are provided for information:

a. Gas filter: MIL−PRF−51525, Drawing 53711-6573660 (Navy), 5-19-6368 (Army). NSN is 4240−01−518−4765.

b. Particulate filter: MIL−PRF−51526, Drawing 6573651 (Navy) and Drawing 5−19−6262 (Army)

c. Filter housing drawings: 5−19−6120, 5−19−6121, and 5−19−12983 (Army) and 53711−6263500, 53711−6573697 (Navy)

6.5 Leakage detector. A halide leak detector manufactured by Nucon Systems, Inc. has been found suitable for this purpose. When using this detector, the presence of high vapor pressure halogenated contaminants in the refrigerant could interfere with the filter leak test resulting in false indications of filter leakage. Should false leakage indications be suspected, the refrigerant should be checked for the presence of such contaminants.

6.6 Adsorbent media. Although manufacturers may consider using alternate adsorbent media, development tests comparable to or beyond those done on ASZM−TEDA carbon per MIL−DTL−32101 would first have to be performed by the contractor and approved for use by the Government. Such testing, which includes gas life capacity testing, using a variety of chemical agents before and after open−air environmental exposure, may be lengthy and costly. All development and test costs would be borne by the contractor.

6.7 Inhalation thresholds. The lower of the 8−hr ACGIH TLVs and OSHA PELs values are extracted from the 4 April 2003 MSDS and summarized in table III. Only current (most recent) values should be used for requirements.

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TABLE III. Inhalation thresholds

Constituent in ASZM−TEDA carbon	Maximum allowed inhalation threshold (mg/m^3)	Cumulative mg in 8 hours
Carbon	5*	13600
Copper	1	2720
Zinc oxide	5	13600
Silver	0.01	27.2
Molybdenum	5	13600

*This is equivalent to a maximum of 13.6 g of carbon collected on a secondary HEPA filter if average airflow is 200 cfm (340 m^3/hr) over 8 hours.

6.8 Shelf life. This specification covers items where shelf life is a consideration. Specific shelf life requirements should be specified in the contract or purchase order. The shelf life codes are contained in the Federal Logistics Information System Total Item Record. Additive information for shelf life management may be obtained from DoD 4140.27−M; Shelflife Management Manual, or the designated shelf life Points of Contact (POC). The POC should be contacted in the following order: (I) the Inventory Control Points (ICPs), and (2) the DoD Service and Agency administrators for the DoD Shelf Life Program. Appropriate POCs for the DoD Shelf Life Program can be contacted through the DoD Shelf Life Management website: http://www.shelflife.hq.dla.mil/.

6.9 Useful life. A useful life goal for the gas filter is 2 years or more of peace time use with no more than 10% degradation in gas life. The useful life is the maximum period of time an unpackaged item can remain fit for use. Exposure to moisture is known to diminish gas life. A DMMP gas life of 60 minutes and challenge of 5000 mg/m^3 is equivalent to 15 chemical attacks, whose average concentration−time is 20,000 mg−min/m^3. Useful life goal for the particulate filter is 4 years or more of peace time use.

6.10 Particulate filter notes.

6.10.1 Filter media. Filter media should have a tensile strength of at least 866 gm/cm (4.85 lb/in) in the machine direction and at least 433 gm/cm (2.42 lb/in) in the cross direction.

6.10.2 Liners. Liners should be used on the inner and outer diameters of the filter media to carry axial loads and protect the media from damage.

6.10.3 Sealant charcteristics. Desirable characteristics of the sealant for securing filtration media to covers are as follows:

a. Tensile strength of at least 91 kg/cm^2

b. Tear strength of at least 28 kg/cm^2.

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c. Adhesion within 5 minutes at 176˚C (350˚F)

6.10.4 Aerosol in test machines. Candidate replacements aerosol materials for potentially hazardous dioctyl phthalate in penetrometer test machines are identified in report CRDEC−TR−271 which is available from Edgewood Chemical Biological Center, AMSRD−ECB−ENE−M, Bldg. E5165, Aberdeen Proving Ground, Maryland 21010−5424. The current type of aerosol used by Edgewood Chemical Biological Center is Durasyn® 164 polyalphaolefin available from Amoco Chemicals, 801 Warrenville Rd, MC 6018, P.O. Box 5206, Lisle, Illinois 60532.

6.11 Subject term (key word) listing.

Collective protection
Filtered air
HEPA filter

Custodian:

Army − EA
Navy − SH
Air Force − 99

Review activities:
Army − MR
DLA − IS

Preparing activity:

Army − EA
(Project No. 4240 −2005−006)

Note: The activities listed above were interested in this document as of the date of this document. Since organizations change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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